

Introduction

While creating scenes with Autodesk MotionBuilder, you can use lighting and shading to introduce a certain degree of visual depth as well as dramatic ambience to your frames. Lights in animation work and behave in the same way as theatrical lights do in the real world. You can place them anywhere in your scene, even inside objects. When done in the right manner, lighting not only adds vitality to your scenes but also acts as a visual guide for your audience. It tells them where to look and the important things to focus on during a scene.

On the other hand, shaders are used to define and modify the surface appearance of 3D models. So, they present you with the scope of designing your objects with the depth and detail that is suitable for your project. Moreover, a shader renders your objects with the descriptive properties of real-time reflectivity or shadowing.

The “Surfaces” feature in Autodesk MotionBuilder allows you to add a dimension of realism to your scenes. The surface appearance of a model simulates the substance from which it is made, which can be anything. For example, an object may be made of cloth, metal, or wood. You can modify the surface appearance of your model with the help of noticeable attributes such as patterns, colours, and shine by applying textures, materials or by adding shadows.

Lights

Custom lights

There are two types of lights under the “Lights” feature in Autodesk MotionBuilder. They are Custom lights and Global lights. There are four types of Custom lights: Point light, Infinite light, Spot light and Area light. When a scene has no light, Global light comes into the picture to provide it with default lighting. You can modify these lights as per your needs and requirements.

Types of Custom lights:

- **Point Light:** It radiates from the centre and illuminates the remaining models that are present in the scene. By using Point lights, you cannot specify the Fog Intensity and the Cone Angle.
- **Infinite Light:** If you want to add light in a specific direction, you can use Infinite light. Here, a downward arrow indicates the colour of the